

Gemini Chat

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [106, 158, 96, 134, 152, 129, 139, 112, 133, 153]

Tap amplitudes (pixels): [216.3631207022121, 213.92522057952868, 215.62931155109686, 219.49487465542333, 209.78083801910984, 203.8872237291979, 212.0, 217.13820483738002, 215.07673049402624, 210.27838690650069, 211.29126815843574]

- ✦ Based on the extracted features provided, here is the analysis and scoring according to the MDS-UPDRS Item 3.4 criteria.

Feature Analysis

1. Amplitude (Excursion)

- **Data:** The pixel amplitudes range from approximately **203 to 219**.
 - Start (Tap 1): 216.36
 - Middle (Tap 6): 203.88
 - End (Tap 11): 211.29
- **Observation:** The amplitude is remarkably stable. There is **no decremending amplitude** (progressive reduction in size). The subject maintains a "large" movement throughout the 11 taps. The variations (e.g., the drop to 203) are negligible and recover immediately in subsequent taps.

2. Speed and Rhythm (Periods)

- **Data:** The periods (in frames) show significant variance, indicating dysrhythmia.
 - Fastest tap: 96 frames
 - Slowest tap: 158 frames

- **Observation:** The rhythm is not metronomic. There is a distinct hesitation early in the sequence (jumping from 106 to **158** frames) which constitutes an **interruption**.
- Following this, the subject speeds up (96), slows down again (134, 152), and fluctuates through the end.
- While exact speed (taps per second) cannot be determined without the video frame rate, the *relative* consistency is poor. The variation between the fastest (96) and slowest (158) represents a ~64% fluctuation, which indicates **slight to mild slowing/hesitation**.

Scoring Rationale

- **Criteria for 0 (Normal):** Rejected. The rhythm is too irregular (96 vs 158 frames) to be considered normal.
- **Criteria for 2 (Mild):** Rejected. While the rhythm is irregular, there are no "3 to 5 interruptions" (there are only 1 or 2 significant spikes in duration: 158 and 152). Crucially, there is **no amplitude decrement**, which is often the deciding factor for higher impairment scores.
- **Criteria for 1 (Slight):** Accepted. The data fits the description of "The regular rhythm is broken with **one or two interruptions or hesitations** of the tapping movement." The spike to 158 frames represents one clear hesitation, and the spike to 152 represents a second. The stable amplitude prevents a higher score.

Final Score

1